S3 Supplementary QC and Robustness

Required negatives, development-gate status, and what would count as robustness

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Short title: S3 Ledger - NDT

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Intent

This supplement holds claim-ledger detail that does not belong in the main theoretical article: closed gates, failed estimator qualifications, and the distinction between synthetic operator existence and biological identification. It is an audit trail, not a second results section.

Required negatives (do not rehabilitate)

The following paths stay closed under current assumptions. A later redesign must say so explicitly; it must not be narrated as a pass.

1. Two-dimensional finite-sample committor recovery is closed. Committor is promoted as a **question**, not as an NMD export.
2. Instantaneous Jacobian recovery does not license ordered finite-time propagation or empirical α/ω via current NMD ingest.
3. Periodic and event-triggered replay/write gates do not beat matched-energy or shuffled-timing controls.
4. Calibrated residual covariance is not memory-specific or target-private reachability.
5. DANDI 000458 EEG-only local-linear operator identification fails (residual-variance fraction 0.856).
6. Repertoire-gate and prestimulus-controlled shape claims on a separate optogenetic resource are closed.
7. A HELS-A neighbourhood robustness extension is not a pass. HELS-A1 fabricated-estimator qualification for a state-matched hippocampal incremental test has not passed family false-positive control.
8. An IBL stimulus-side population decoder can pass a pilot positive-control test while a predefined single-trial discrete chart-transition operator fails its held-out residual floor (median residual fraction ≈ 1.001). Failure of that readout is not a neural-information null.
9. A three-stage meso $\rightarrow$ regional $\rightarrow$ surface-macro prediction chain on DANDI 000458 is blocked: event alignment remains inconclusive and the parent feature table is not region- or unit-resolved. The parent claim-class-1 associations are neither promoted nor demoted by that gate.
10. A joint TRN $\rightarrow$ mesoscopic $\rightarrow$ surface-macro causal bridge is not estimable from the audited public archives. DANDI 000458 has meso and surface EEG without a usable TRN perturbation; the companion Figshare resource has timed TRN perturbation without surface EEG. Pooling the two does not identify the missing chain.
11. Condition mixing can inflate the innovation term of a local-linear predictive-spread fit. In a companion Allen analysis, pooling expected and omission epochs increased $\mathrm{Tr}(Q)$ in all 126 examined session $\times$ region combinations, typically several-fold. Residual degrees of freedom were 1 throughout those six-epoch fits of A, Q, and $\Sigma^{(h)}$. Condition-separated fitting is required before any predictive-spread interpretation. Chart-speed coupling of regional trajectories is not stochastic short-horizon reachability.

Rhythmic variance control

Scalar $r(t) \rightarrow \sigma(t)$ remains the **minimal** special case of accessibility control. Event-locked spindle d_n timing is a chart constraint (S2), not causal denoising. Write-gate specificity against energy-matched controls is not supported. Power-matched and phase-shuffle nulls remain unpaid for the full discriminant.

Thalamic access

Two-operator existence on a frozen landscape is computational (Langell, 2026e). In one auditory TRN-to-relay preparation, PV-selective photoinhibition is associated with evoked participation-ratio contraction (Langell, 2026f); a full TRN variance-gate hypothesis, including a scalar Fano clamp, is not validated. On the same resource, both PV- and SST-family photoinhibition associate with non-scalar change in residual covariance **shape** after prestimulus cleaning (Langell, 2026g) — a bounded contrast against scalar rescaling, not identification of A_{noise} or of a chart Jacobian. Stronger repertoire-

gating and prestimulus-controlled shape claims on a separate optogenetic resource have been rejected. Direct mapping onto NDT operators remains open. A joint TRN → meso → surface-macro test is not currently estimable from the audited archives (item 10). NDT has not found the thalamic mechanism.

HELS versus HELS-A versus HELS-A1

- **HELS (legacy):** where does V come from? Infrastructure is not writing.
- **HELS-A:** context operator A_H on frozen V. Synthetic access can change first-passage and unsaturated committor bias without rewriting the landscape (Langell, 2026h). Official confirmatory seeds were not consumed as a pass. A neighbourhood robustness gate failed.
- **HELS-A1:** header eligibility of a CA1–prefrontal archive is not a hippocampal mechanism. Spike-time scoring remains closed until an estimator is independently qualified. Header yield is not a CA1 result.

A measured-state history constraint (recent history improves linear prediction of subsequent EEG beyond current state on DANDI 000458) is P6-adjacent. It is not landscape writing. The corresponding prediction uses measured chart state x_t, not latent X_t.

Dendritic mobility bias

A companion simulator separates fast dendritic access from a slower phenomenological bias M_b (Langell, 2026i). The same logical move — control changing futures without rewriting content — can be stated at the dendrite. That is a speculative microscale analogue, not an empirical MNPS identity, and the micro–meso–macro chain remains mostly dashed.

Robustness meaning by layer

Layer	What would count as robustness
State chart	Same direction under alternative features, preprocessing, and site
Local J / a	Truth-known envelopes **and** residual-quality gates on the empirical chart
Reachability	Horizon/noise sensitivity plus a residual gate; not Gramian controllability
Committor / FAR	Declared destinations / perturbation family; currently conceptual
Mechanism	Causal or discriminative test against arousal, power, and matched-energy controls

Table 1: Robustness is layer-specific. Surviving a clinical contrast is not operator identification.

Robustness summary

The article's discipline is negative as much as positive. Spindle timing replicates on two datasets and is incomplete on matched-control specificity. Coma geometry is recovery-linked on the invalidity-fixed 301-patient merge (S2). Meso–macro contact does not license a different inferential claim class. IBL decoder PASS versus discrete-flow STOP is the same lesson on a different chart. Closed gates stay closed.

Bibliography

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